

ReadySetBag!

An Emergency Go-Bag Preparedness Game for Android



1. App Concept Description (Abstract)

ReadySetBag! is a 3D mobile serious game for Android that teaches students how to prepare an emergency go-bag. Racing against a countdown, the player explores a house, searches drawers, cabinets and shelves for supplies, and packs a backpack with a strict 5 kg weight limit, choosing life-saving essentials over tempting but useless items. Once the bag is packed, a teacher character presents disaster scenarios in Filipino, such as an earthquake blackout or being trapped behind a collapsed wall, and the player answers by dragging the correct item out of the bag. Each correct answer opens a short hands-on minigame that shows how the item is actually used. A 100-point score rates packing, quiz, task and time performance. Teachers can run live class sessions through a join code with results saved to the cloud, while offline practice lets students replay and unlock harder levels.

2. Hardware and Software Requirements

2.1 End-User Device (to play the game)

Component	Minimum	Recommended
Operating system	Android 7.1 (API level 25)	Android 10 or newer
Processor	64-bit ARM (arm64-v8a) processor	Octa-core 64-bit ARM, 2.0 GHz+
Graphics	OpenGL ES 3.0 or Vulkan capable GPU	Vulkan capable GPU
Memory (RAM)	3 GB	4 GB or more
Storage	About 250 MB free (installer is about 67 MB)	500 MB free
Display	Touchscreen, landscape orientation	6 inch or larger, 1080p
Network	None for Practice mode	Internet (Wi-Fi or mobile data) for student log in and Teacher Sessions
Other	Speakers or headphones; vibration motor (optional)	

2.2 Development Environment

Item	Specification
Development PC	Windows 10/11 64-bit, quad-core CPU, 16 GB RAM, DirectX 11/12 GPU, 30 GB free disk space
Game engine	Unity 6 (6000.3.24f1) with the Universal Render Pipeline (URP)
Programming language	C# (.NET), IL2CPP scripting backend
IDE	Microsoft Visual Studio
Build target	Android SDK and NDK via Unity Android Build Support; OpenJDK; Android Studio emulator / physical device for testing
Unity packages	Universal RP, Cinemachine, TextMeshPro, uGUI, Video Player, 2D Sprite, Unity Test Framework
Backend services	Google Firebase: Authentication (student accounts) and Cloud Firestore (sessions and results)
Art and assets	Aseprite (pixel-art sprites and UI), 3D house models (FBX), Jersey 25 font
Version control	Git

3. Product Features

- **Explore-and-pack gameplay.** Move through a fully modelled 3D house with an on-screen joystick and camera controls. Supplies are hidden in furniture; tapping a drawer or cabinet opens it so its contents can be dragged into the go-bag.
- **Grid-based go-bag inventory.** Items take up real space on a grid and have real weights. Three bag types (small, medium and standard backpack) each have their own pockets, and the bag cannot exceed 5 kg.
- **41 supply items in five importance tiers.** Critical (e.g. water bottle, first aid kit, flashlight, whistle), Important, Useful, Conditional and Nuisance items (e.g. game console, pillow, soy sauce) teach students to tell essentials from clutter.
- **Scenario quiz answered with the bag.** A bank of 20 disaster-scenario questions written in Filipino. Players answer by dragging an item from their own bag, so an essential left behind cannot be used to answer. Each answer is followed by an explanation.
- **20 hands-on minigames.** One for every quiz question. Each correct answer opens a quick activity showing how the item is used: shining a flashlight or glow stick to find people, blowing a whistle for rescuers, dressing a wound, pouring water, tuning a radio, loading batteries, tying a rope knot, and more.
- **Three difficulty levels with unlocks.** Beginner (10 min, 10 questions), Intermediate (8 min, 15 questions, garage opens) and Advanced (6 min, 20 questions, second floor opens). Scoring 70+ unlocks Intermediate and 85+ unlocks Advanced.
- **100-point drill score and badges.** Score = 40% packing + 20% quiz + 20% minigame completion + 20% time bonus. Results show a breakdown and a learning-stage badge: Cognitive, Associative or Autonomous.
- **Teacher sessions and student accounts.** Students log in with Firebase accounts and join a teacher's live session with a code. The teacher sets the difficulty, and each student's result is uploaded to Cloud Firestore for the teacher/admin dashboards (saved on the device and re-sent if offline).
- **Offline practice mode.** Students (or guests) can play anytime without internet; scores, unlocked levels and the Journal are saved on the device.
- **Item Journal.** Every item a player has packed is unlocked in a Journal that shows its weight, importance and a short description, building a personal reference of go-bag supplies.
- **Replayability.** Player and go-bag spawn points are randomized each run and questions are drawn from the bank in random order, so every drill is different.
- **Polish and accessibility.** Selectable characters, pixel-art UI, guided tutorial, music and sound effects with volume settings, haptic feedback, pause menu, and resume support if the game is closed mid-drill.

4. Screenshots and Descriptions



Figure 1. Title Screen

The game's pixel-art logo. Tapping anywhere continues to the login screen.

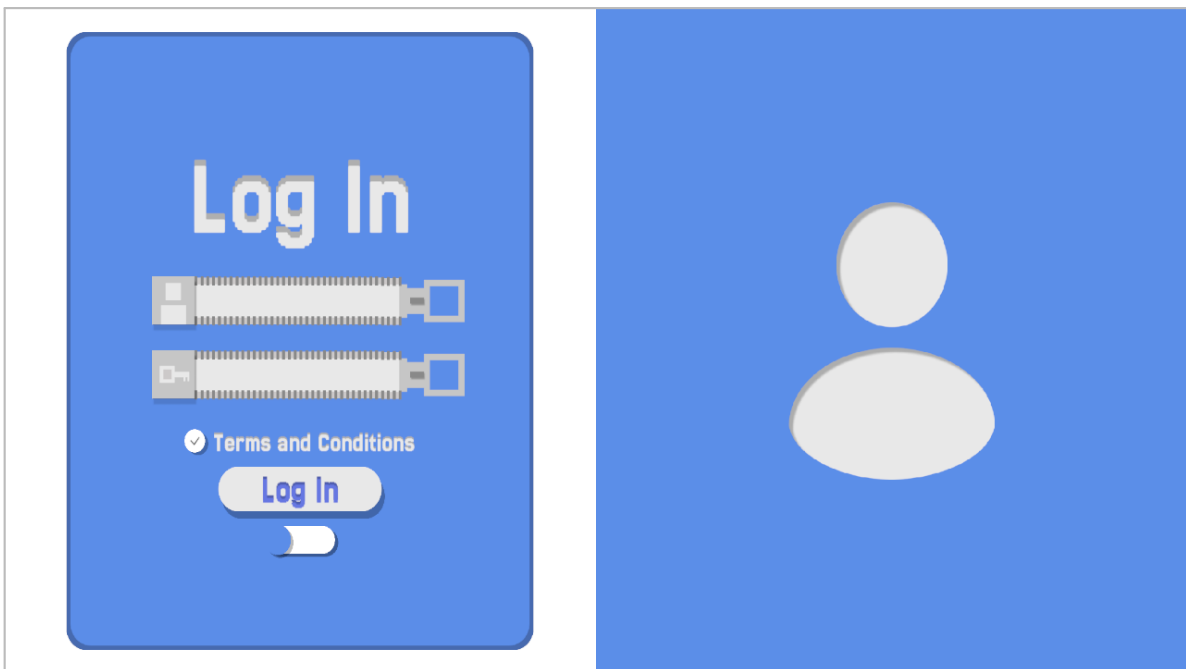


Figure 2. Student Log In

Students sign in with the account their teacher created. A toggle switches to guest play, and the Terms and Conditions must be accepted before playing.



Figure 3. Main Menu

Play, Options, Switch (character select), Exit, How to Play and About, over an animated bedroom scene. The ID card shows the logged-in player.

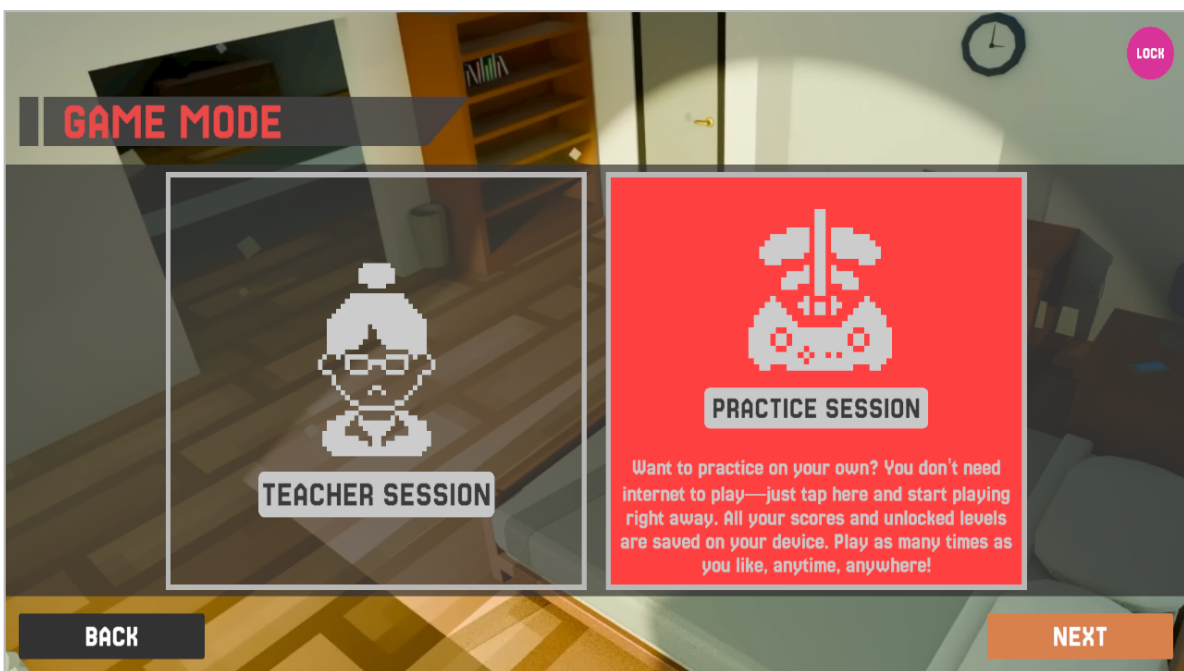


Figure 4. Game Mode Selection

Teacher Session joins a live class session by code; Practice Session lets the student play offline as many times as they like.

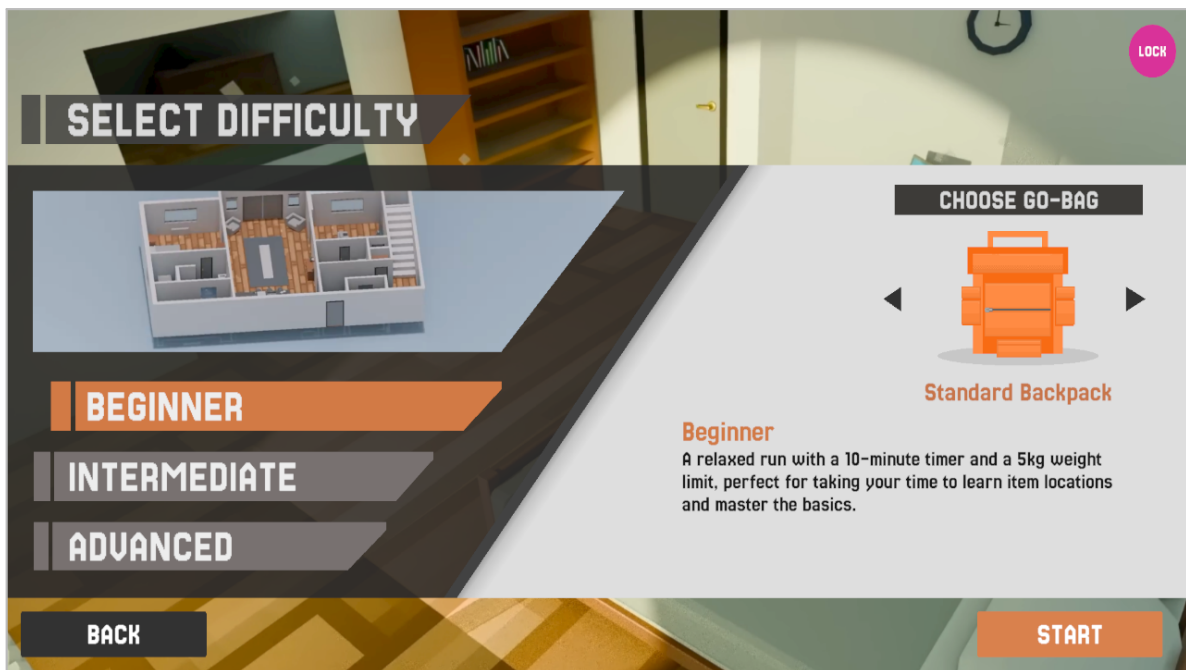


Figure 5. Difficulty and Go-Bag Selection

Choose Beginner, Intermediate or Advanced (each with a map preview and rules) and pick which backpack to carry.

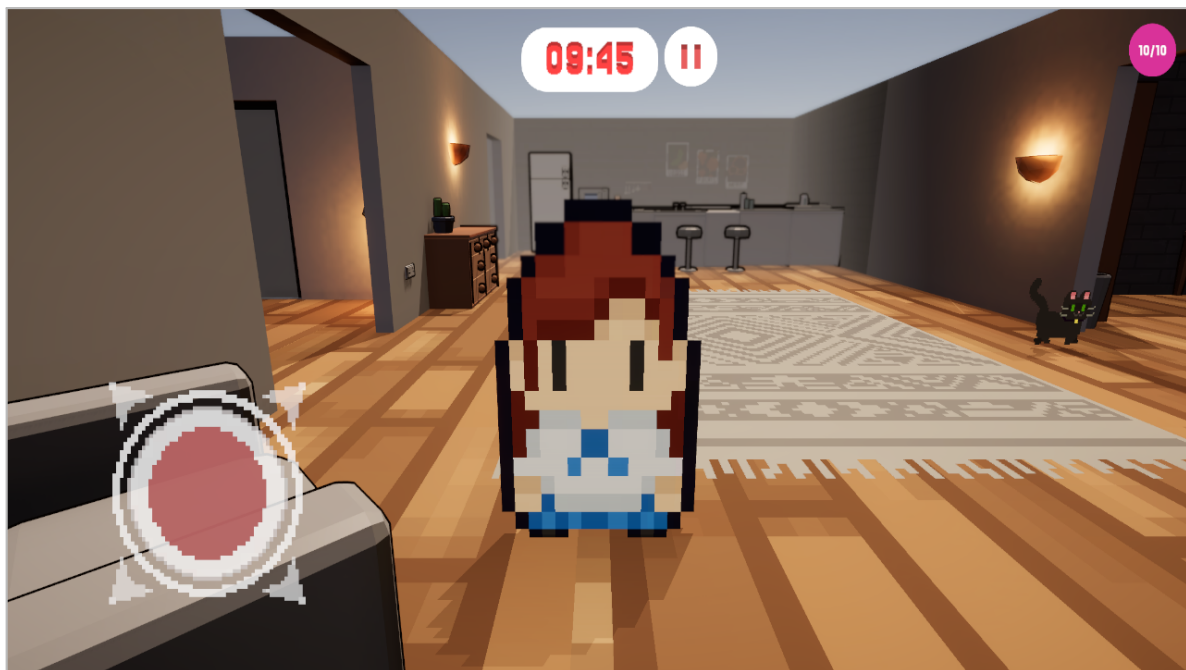


Figure 6. Exploring the House

The player moves with the virtual joystick. The countdown timer is shown at the top; the house includes details such as the family cat wandering around.



Figure 7. Searching Storage and Packing

Opening furniture shows its compartments (right) next to the go-bag grid (left). Items are dragged into the bag's pockets; the bag icon shows how full it is by weight.



Figure 8. Item Journal

Unlocked items with their weight, importance level and description. Locked slots are items the player has not packed yet.



Figure 9. Scenario Quiz

The teacher character presents an emergency scenario (here: a blackout after an earthquake). The player drags the correct item from the bag into the answer box.



Figure 10. Answer Feedback

After answering, the teacher explains why the item is right (here: a whistle carries farther than a voice and does not tire you out while waiting for rescue).

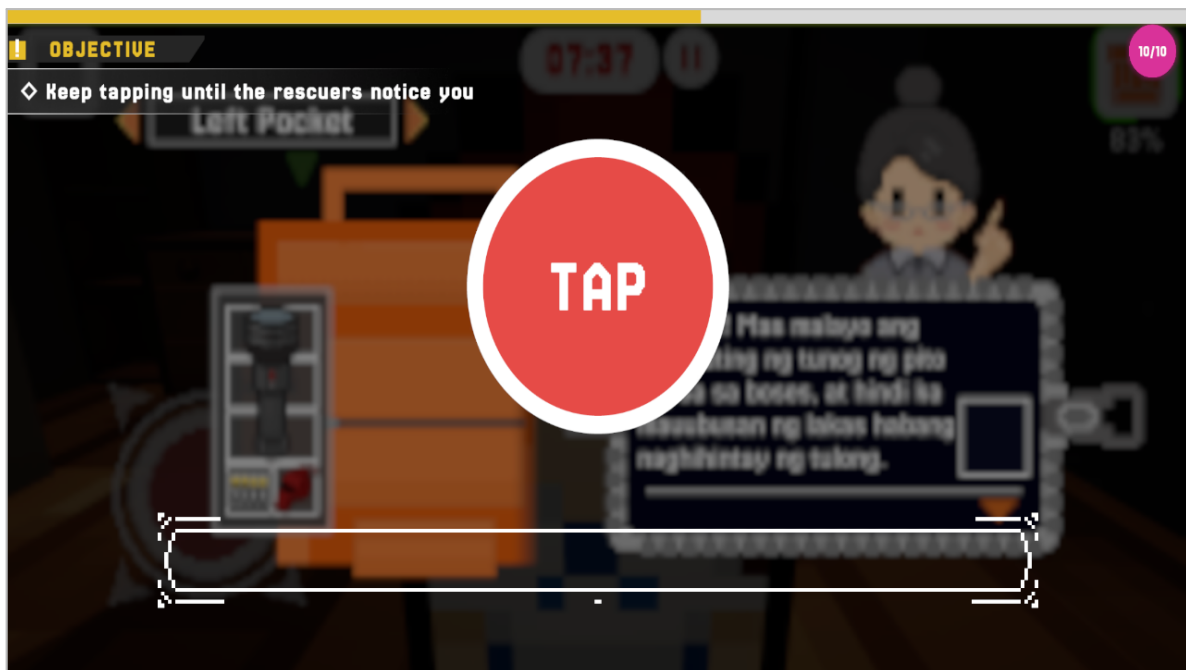


Figure 11. Minigame: Whistle

A correct answer opens a timed minigame. Here the player taps repeatedly until the rescuers notice them.



Figure 12. Minigame: First Aid

The player checks the arm for wounds, cleans the bruises and covers the wound with gauze using tools from the first aid kit.



Figure 13. Results

The final drill score with a breakdown of packing, quiz, minigame and time bonus, plus the learning-stage badge. Shown from a sample test run; from here the player can retry, move to the next difficulty or return to the menu.